

Spatial Patterns in Educational Attainment

Research Progress Report | Week 8

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1 National BA+ Trajectory, 1940-2024

The simplest thing to look at first in the long panel is the national trajectory itself. Figure 1 plots the population-weighted BA+ share of adults 25 and over at each of the 18 anchor years the panel covers. The share rose from 4.6% in 1940 to 35.7% in 2024, a roughly eightfold increase over 84 years.

The interesting part is not the trend itself but where the growth was fast and where it was slow. Table 1 breaks the trajectory into the average growth per year between consecutive anchor points. The 1940-1970 period grew at only 0.20 pp per year, which is the slowest sustained rate in the panel. The 1970s more than doubled that pace at 0.56 pp per year, and this decade holds the fastest sustained rate anywhere in the series. Growth slowed again to about 0.41 pp per year through the 1980s and 1990s, dipped further to 0.34 through the early 2010s, then accelerated sharply in the last five years to 0.72 pp per year, the fastest sustained rate since the 1970s.

So there are two acceleration episodes in the whole series: the 1970s (GI Bill effects and the mass expansion of higher education after Vietnam) and the post-2019 window. Everything in between grew more slowly than either of those bursts. This is a cleaner story than "growth sped up after 2000" would have been, since 2000-2012 was actually the slowest stretch since 1970.

Period	Start (%)	End (%)	Total change (pp)	Growth (pp/yr)
1940-1970	4.6	10.7	6.1	0.20
1970-1980	10.7	16.2	5.6	0.56
1980-1990	16.2	20.3	4.1	0.41
1990-2000	20.3	24.4	4.1	0.41
2000-2012	24.4	28.5	4.1	0.34
2012-2019	28.5	32.1	3.7	0.52
2019-2024	32.1	35.7	3.6	0.72

Table 1: National BA+ share at anchor years and growth per year between them. The 1970s and the post-2019 window are the two acceleration episodes; everything in between grew more slowly.

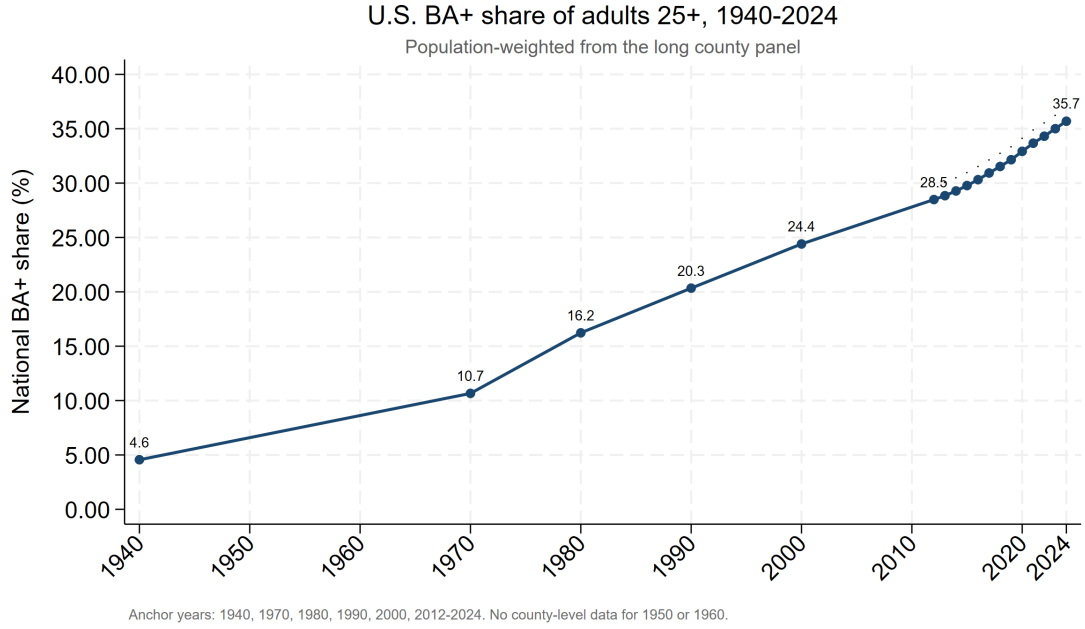


Figure 1: National BA+ share of adults 25 and over, 1940-2024, from the population-weighted long county panel. The 1970s and the post-2019 window are the fastest-growing periods.

File location

The trajectory table is at `output/tables/natl_baplus_trajectory.csv` and the figure is at `output/figures/natl_baplus_trajectory.png`. The build script is `code/24_natl_baplus_trajectory.do`.

2 Dispersion of BA+ across counties, 1940-2024

The second question we can answer using the long panel is how the spread of BA+ shares across counties has changed over time. There are two measures we can think of: the raw gap between the 90th and 10th percentile counties (P90-P10), which shows the spread in percentage points, and the coefficient of variation ($CV = \text{standard deviation} / \text{mean}$), which shows the spread relative to the average. Both need to be looked at together because they can move in opposite directions.

Table 2 shows both measures at each decadal anchor year. The story is that absolute divergence and relative convergence happened at the same time. The P90-P10 gap grew from 3.7 pp in 1940 to 24.5 pp in 2024, a nearly sevenfold increase. Every anchor year is wider than the one before, with no interruption. Meanwhile the CV fell from 0.52 to 0.42 over the same 84 years. The mean of the distribution rose so fast that the relative spread actually compressed even as the raw gap kept expanding.

One thing that is remarkable is 1940 itself. The CV of 0.52 was not dramatically higher than any later period. Even when almost no one had a BA (national mean was 3.3%), the relative spread across counties was already close to today’s level. Spatial inequality in human capital has always been substantial in relative terms. The other detail worth noting is the 1970s. That was the only decade in the panel where the CV rose (from 0.52 to 0.54), which happened at the same time national growth was fastest. Growth in the 1970s was not distributed evenly since some counties raced ahead while others did not keep up.

Year	P90 - P10 gap (pp)	CV (SD / Mean)	National mean BA+ (%)
1940	3.7	0.52	3.3
1970	8.0	0.54	7.4
1980	11.6	0.48	11.5
1990	14.4	0.49	13.5
2000	17.4	0.47	16.5
2012	20.3	0.45	19.5
2024	24.5	0.42	24.7

Table 2: Absolute (P90-P10) and relative (CV) dispersion of county BA+ shares, 1940-2024. Absolute gap grew sevenfold while the relative gap shrank. Each county counts once (unweighted).

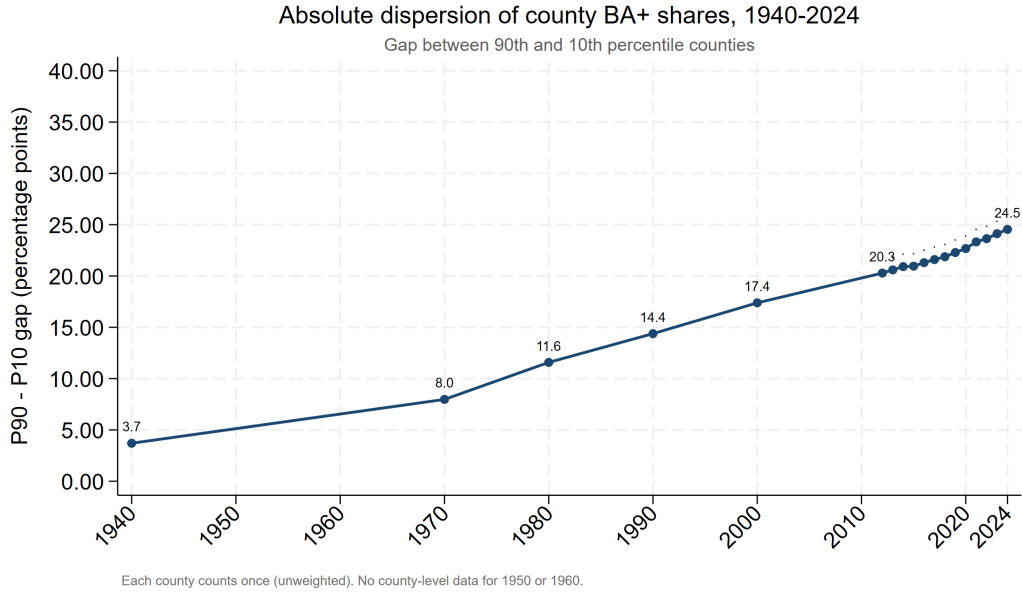


Figure 2: P90 - P10 gap across U.S. counties, 1940-2024. Every decade is wider than the one before. No pause in the divergence.

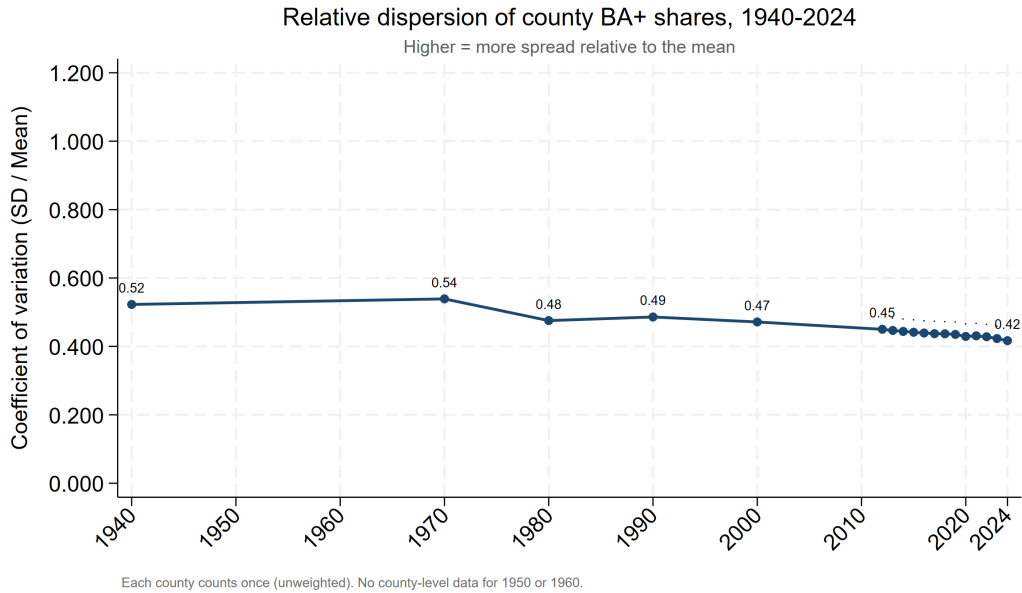


Figure 3: Coefficient of variation across U.S. counties, 1940-2024. The 1970s are the only decade with rising relative spread. Everything else shows gradual compression.

The absolute picture (Figure 2) says spatial inequality got worse every decade. The relative picture (Figure 3) says the spread of BA+ shares shrank relative to the mean, from 0.52 to 0.42,

which means the country as a whole rose faster than the spread expanded. This is the same tension that shows up in the tract-level analysis from earlier reports, now confirmed at the county level over a much longer window.

File location

The dispersion table is at `output/tables/dispersion_over_time.csv`. The figures are at `output/figures/dispersion_p90_p10.png` and `output/figures/dispersion_cv.png`. The build script is `code/25_dispersion_over_time.do`.

3 Rank Persistence in County BA+ Share, 1940 vs 2024

The next question is whether the counties on top in 1940 are still on top in 2024. If yes, geography of human capital in America is deeply persistent. If no, 84 years of migration, industry change, and higher-education expansion have scrambled the map. One of the ways to measure this is to rank every county by BA+ share in 1940, rank them again in 2024, then check how similar the two rankings are. Spearman rank correlation puts a single number on it, running from -1 (perfect reversal) through 0 (no relationship) to 1 (perfect persistence).

The Spearman correlation is 0.512 ($p < 0.001$), and Kendall's tau (a more conservative rank measure) is 0.359. Both are statistically clean, so the persistence is visible. But 0.512 is far from 1.0. That means the 1940 ranking predicts only about half of the ordering in 2024, which is a middle-ground result: neither "same places on top forever" nor "everything reshuffled." Figure 4 shows this visually. Each dot is a county. If persistence were perfect, all 3,088 dots would be on the dashed 45-degree line. They do not. But there is clear clustering along the diagonal, especially in the bottom-left corner (counties low in 1940 that are still low in 2024) and the top-right corner (elite counties that stayed elite).

The more interesting finding comes from looking at the corners of the distribution separately. Counties in the bottom 10% in 1940 had a 40% chance of still being in the bottom 10% in 2024 (122 of 309 counties). Counties in the top 10% in 1940 had a 32% chance of still being in the top 10% (100 of 309). Both numbers are far above the 10% you would expect by chance, so persistence is strong at both ends. But being at the bottom is stickier than being at the top. Elite counties in

1940 got displaced more often than poor counties escaped. Table 3 summarizes.

Measure	Value	Interpretation
Spearman correlation	0.512	Moderate persistence
Kendall's tau	0.359	Moderate persistence (conservative)
Bottom 10% persistence	39.5%	Stuck at bottom over 84 years
Top 10% persistence	32.4%	Stayed on top over 84 years
Counties in both years	3,088	Panel used for the calculation

Table 3: Rank persistence in county BA+ share between 1940 and 2024. Persistence is real at both ends of the distribution but stronger at the bottom.

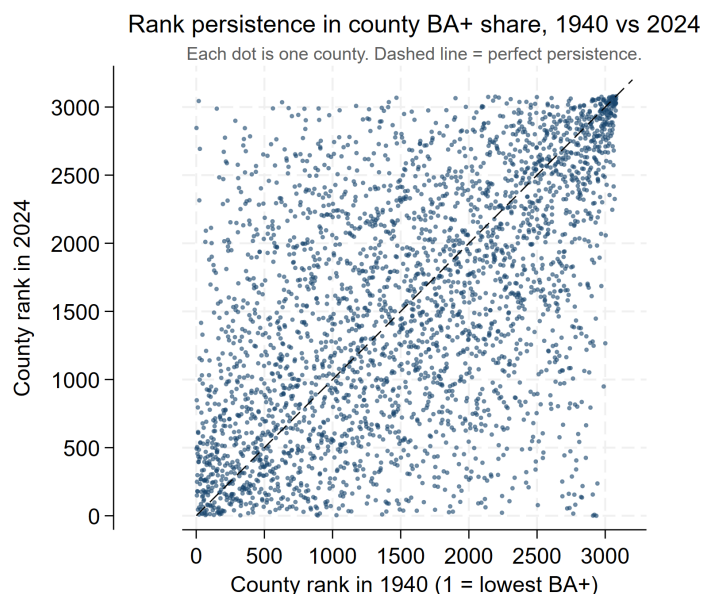


Figure 4: Each county's rank in 1940 (x-axis) versus its rank in 2024 (y-axis). Dashed line is perfect persistence. Substantial clustering along the diagonal shows real persistence, but the wide scatter shows that many counties moved a lot.

The counties that climbed the most were not university towns or lab towns; they were suburban and exurban counties in Sun Belt metros. Georgia dominates the list, with Forsyth, Fayette, Dawson, Cherokee, and Columbia counties all appearing in the top ten. These are Atlanta-area exurbs that transformed from small rural counties in 1940 into affluent bedroom communities by 2024. Forsyth is the extreme case, jumping from a BA+ share of 0.77% in 1940 (a rank near the very bottom) to 57.1% in 2024 (near the very top), a 3,027-position climb. Other climbers

follow the same pattern: Fort Bend (Houston suburbs), Stafford (D.C. exurbs), Monroe IL (St. Louis exurbs), and Fayette GA (Atlanta suburbs). The mechanism is not that these counties built universities; it is that they absorbed college-educated migrants from major metros as those metros sprawled outward. Table 4 shows the top ten.

County	State	1940 BA+ (%)	2024 BA+ (%)
Forsyth	Georgia	0.77	57.13
Keweenaw	Michigan	0.40	41.19
Fayette	Georgia	1.24	49.90
Dawson	Georgia	0.81	36.44
Stafford	Virginia	1.51	43.83
Monroe	Illinois	1.38	38.67
Cherokee	Georgia	1.64	41.15
Campbell	South Dakota	1.36	36.03
Fort Bend	Texas	1.93	50.23
Columbia	Georgia	1.63	38.46

Table 4: Ten counties that climbed the most in rank between 1940 and 2024. Suburban and exurban growth in Sun Belt metros dominates the list.

The counties that fell the most tell a different story. They are almost all rural West Texas and Nevada counties with economies built on oil, ranching, or mining. Every county in the top-ten fallers is in either Texas (8 of 10) or Nevada (2 of 10). Their 1940 BA+ shares were in the 5-6% range, which was above the national mean of 4.6% and put them in respectable middle-of-the-pack positions. But as the country rose to a national BA+ share of 35.7%, these counties barely moved. Their 2024 shares are still in single digits or low teens. Loving County, Texas is the extreme case with a 2024 BA+ share of 0.00%, though this is partly an artifact of the county having only about 60 residents; even a small number of college-educated adults would show up as a nonzero share, and the fact that it does not means every adult with a BA has left. The pattern for the rest of the list is real: extractive-industry towns that never diversified and slowly emptied. Table 5 shows the pattern.

County	State	1940 BA+ (%)	2024 BA+ (%)
Loving	Texas	6.33	0.00
Reeves	Texas	6.11	7.40
Pershing	Nevada	6.00	6.15
Crane	Texas	5.99	12.61
Upton	Texas	5.20	8.71
Reagan	Texas	6.40	13.59
Deaf Smith	Texas	5.27	11.43
Moore	Texas	5.53	12.30
White Pine	Nevada	5.48	12.36
Crockett	Texas	5.13	10.68

Table 5: Ten counties that fell the most in rank between 1940 and 2024. Rural West Texas and Nevada mining/oil counties dominate the list.

Two things from the fallers are noticeable. First, the raw BA+ shares in these counties did rise between 1940 and 2024, sometimes tripling or quadrupling. But the country as a whole rose so much faster that these counties fell dramatically in the ranking. Second, most started ABOVE the national mean of 4.6% in 1940. So we are not watching counties that were always poor; we are watching counties that were solid middle-of-the-pack in 1940 and got left behind as the national distribution shifted upward. The persistence story is not "the geography of poverty stays the same." It is more specific: certain regions (rural Great Plains and Appalachia) failed to keep pace with a national trend that reached almost everywhere else.

File location

The full list of counties with their rank changes is at `output/tables/rank_persistence_movers.csv`. The scatter plot is at `output/figures/rank_persistence_scatter.png`. The build script is `code/26_rank_persistence_1940_2024.do`.

4 Metro-Rural BA+ Gap Over 84 Years

The next step would be to see whether the metro-rural gap in BA+ attainment is a modern phenomenon or something that has existed for as long as we can measure. Using RUCC to classify each county as Metro (1-3), Nonmetro with city (4-7), or Rural (8-9), we can compute the population-weighted BA+ share for each group at every anchor year from 1940 to 2024. Note: RUCC editions only start in 1974, so the 1940 and 1970 values match to the 1974 edition, which is an imperfect but standard approximation.

The answer is that the gap has always been there. In 1940, Metro counties had a BA+ share of 4.9% while Rural counties had 2.3%, a gap of only 2.6 percentage points but a ratio of more than two to one. By 2024 the gap had grown to 18.1 percentage points, with Metro at 37.8% and Rural at 19.6%. Table 6 shows the trajectory at each decadal anchor point.

Year	Metro (%)	Nonmetro with city (%)	Rural (%)	Metro-Rural gap (pp)
1940	4.9	3.3	2.3	2.6
1970	11.8	7.7	5.3	6.4
1980	17.6	11.2	9.0	8.6
1990	19.0	13.1	10.1	8.9
2000	26.4	15.3	13.0	13.4
2012	30.5	17.8	15.6	14.9
2024	37.8	23.6	19.6	18.1

Table 6: Population-weighted BA+ share by RUCC-based metro status, 1940-2024. The metro-rural gap grew every decade with no exception.

Figure 5 shows all three lines together. Two things are worth pointing out. First, the ordering of the three groups is identical at every year in the panel: Metro is always highest, Rural is always lowest, and Nonmetro-with-city is in between. This ordering has never flipped in 84 years. Second, the three lines fan out over time. In 1940 they were close together (spread of 2.6 pp between top and bottom). By 2024 they are far apart (spread of 18.1 pp). This is the same absolute-divergence story that we showed for the P90-P10 gap, but now with an economic interpretation attached to it: the divergence has an urban-rural dimension.

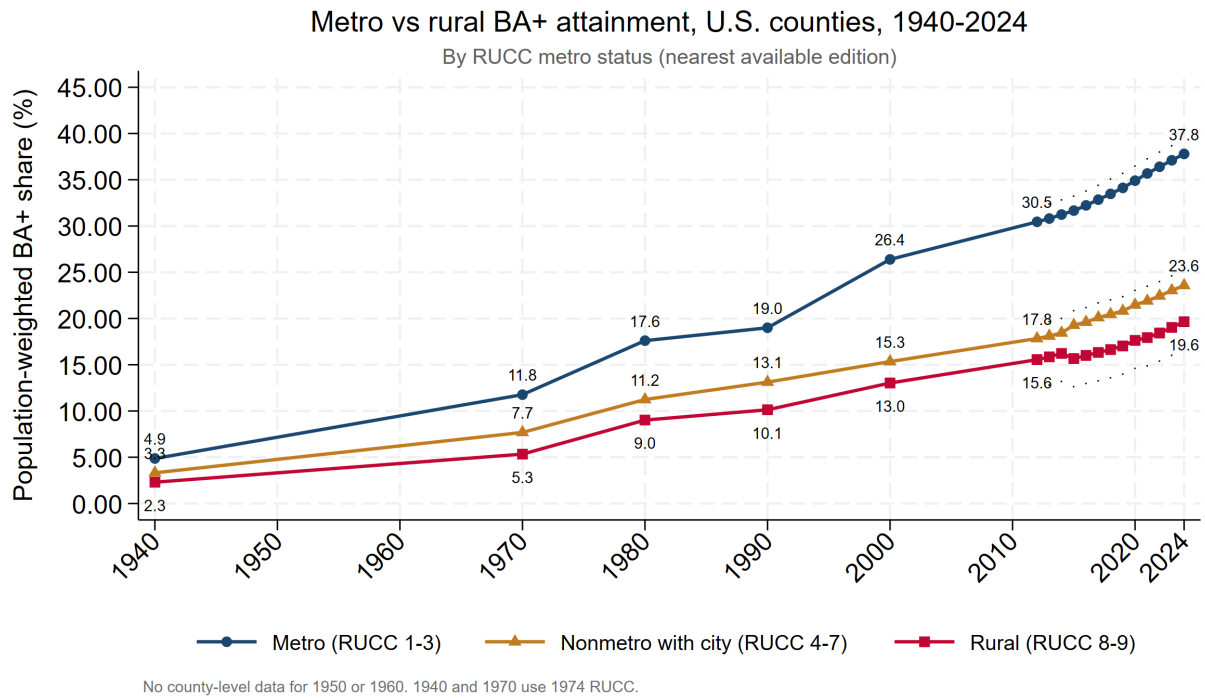


Figure 5: Population-weighted BA+ share by RUCC metro status, 1940-2024. The three groups keep the same ordering across all 84 years, but the spread between them grows steadily.

Figure 6 isolates just the Metro-Rural gap. The chart shows two clear acceleration episodes. The first is 1990-2000, when the gap widened by 4.5 percentage points in a single decade (from 8.9 to 13.4). This is the biggest single-decade jump anywhere in the panel and lines up with the college wage premium literature that identifies the 1990s as the takeoff period for skill-biased returns. The second episode is 2000-2012, when the gap widened another 1.5 percentage points. After 2012 the gap continues to widen at about 0.3 percentage points per year, slower than the 1990s but still steady.

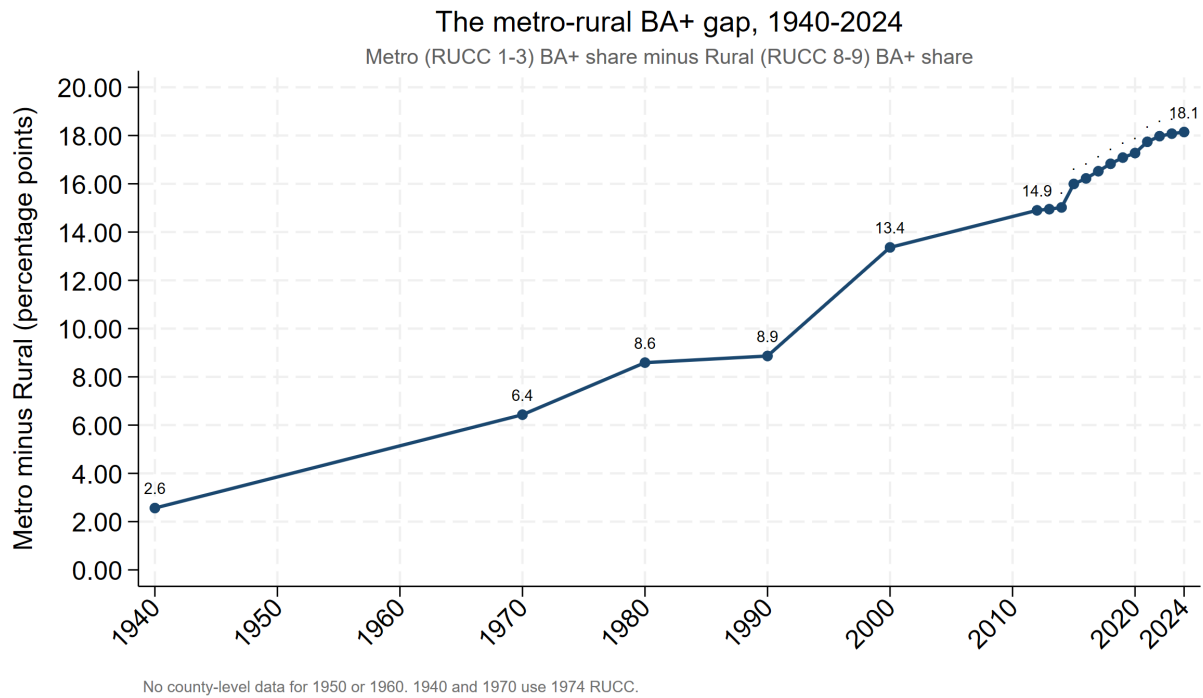


Figure 6: Metro (RUCC 1-3) BA+ share minus Rural (RUCC 8-9) BA+ share, 1940-2024. Two clear acceleration episodes: 1990-2000 and 2000-2012. The gap never narrows in any decade.

From this observation, the takeaway is that the metro-rural attainment gap has existed for at least 84 years and has never narrowed in any decade of the panel. What has changed is the size of the gap, which grew from 2.6 percentage points to 18.1 over 84 years. The 1990s were the sharpest acceleration decade. This connects directly back to our absolute divergence finding: the widening dispersion of county BA+ shares is not random, it has a specific geographic character, and that character is metropolitan concentration of college-educated adults.

File location

The trajectory table is at `output/tables/metro_rural_trajectory.csv`. The figures are at `output/figures/metro_rural_shares.png` and `output/figures/metro_rural_gap.png`.

The build script is `code/28_metro_rural_gap_over_time.do`.

5 Kentucky in the National Context

Kentucky is the state this project is embedded in, so it is worth placing it explicitly in the national picture the panel has been describing. We want to know 1) how does Kentucky compare to the national average across time, 2) which counties gained and lost the most, and 3) how has Kentucky’s position in the national bottom tier changed.

The state has fallen further behind the U.S. average every single decade in the panel, but this hides a more nuanced story about the county-level distribution. Table 7 shows Kentucky’s population-weighted BA+ share against the national share at each anchor year. In 1940, Kentucky was 1.6 pp behind the country (2.9% versus 4.6%). By 2024, the gap had grown to 8.1 pp (27.6% versus 35.7%). This is monotonic: every anchor year shows the state further behind than the year before it.

Year	Kentucky (%)	United States (%)	Gap (pp)
1940	2.95	4.56	-1.61
1970	7.21	10.66	-3.45
1980	11.08	16.23	-5.15
1990	13.63	20.34	-6.71
2000	17.14	24.40	-7.27
2012	20.98	28.49	-7.51
2024	27.55	35.69	-8.14

Table 7: Population-weighted BA+ share for Kentucky and the United States, 1940-2024. The gap grew from 1.6 pp to 8.1 pp over 84 years.

Figure 7 shows the two trajectories side by side. Kentucky grew at similar rates to the country in each decade but starts from a lower base, so the two lines never converge; they run roughly parallel while getting further apart in absolute terms.

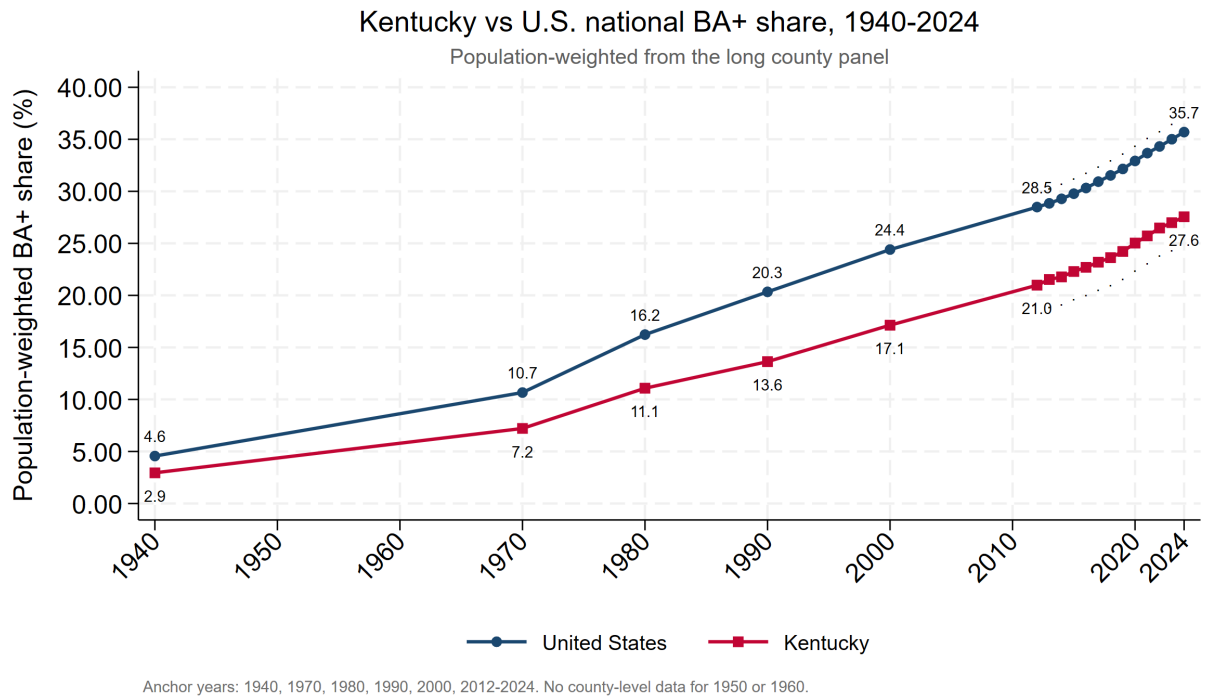


Figure 7: Kentucky and U.S. national BA+ share, 1940-2024. Kentucky consistently trails the country. The absolute gap widened from 1.6 pp in 1940 to 8.1 pp in 2024.

The state-level average hides big differences across counties. Table 8 shows Kentucky’s eight counties with the largest 1940-to-2024 BA+ gain, and every single one is a Golden Triangle county: Lexington (Fayette), Louisville (Jefferson), Northern Kentucky (Boone and Kenton), and their affluent bedroom communities (Oldham, Woodford, Scott). Fayette gained 40 pp, Oldham gained 44 pp. These counties look nothing like the falling-behind state average.

County	1940 BA+ (%)	2024 BA+ (%)	Gain (pp)
Oldham	2.36	46.42	44.06
Fayette	8.14	48.29	40.15
Campbell	3.22	39.74	36.52
Woodford	5.41	40.74	35.34
Kenton	3.06	37.43	34.37
Boone	2.56	35.71	33.15
Jefferson	4.61	37.03	32.42
Scott	5.10	34.73	29.64

Table 8: The eight Kentucky counties with the largest BA+ gain, 1940 to 2024.

Table 9 shows the opposite end. These are the eight counties with the smallest gains, and every one is in Eastern Kentucky, southern Kentucky, or an isolated rural county. Owsley gained 6.121 pp over 84 years, and its 2024 BA+ share of 7.1% is still lower than the U.S. national mean was in 1970 (10.7%). nicholas, Wolfe, McCreary, and Elliott are all in the same category: counties whose 2024 BA+ shares would have looked normal in 1970 but are far behind in 2024.

County	1940 BA+ (%)	2024 BA+ (%)	Gain (pp)
Owsley	1.00	7.12	6.12
Nicholas	2.52	9.82	7.30
Wolfe	1.31	8.69	7.37
McCreary	1.01	8.48	7.47
Elliott	0.91	8.67	7.77
Jackson	0.89	8.84	7.95
Lee	1.29	9.25	7.96
Butler	0.98	9.87	8.89

Table 9: The eight Kentucky counties with the smallest BA+ gain, 1940 to 2024. Concentrated in Eastern Kentucky and rural areas.

The most interesting piece of the picture is Table 10, which counts the number of Kentucky counties that are in the national bottom 10% of the BA+ distribution at each anchor year. In

1940, 53 of Kentucky’s 120 counties (44.2%) were in the national bottom decile. By 2024, only 29 counties (24.2%) are. The count nearly halved over 84 years, most of the drop happening between 2012 and 2024.

Year	KY counties in bottom 10%	Total KY counties	Share of KY (%)
1940	53	120	44.2
1970	45	120	37.5
1980	55	120	45.8
1990	55	120	45.8
2000	55	120	45.8
2012	45	120	37.5
2024	29	120	24.2

Table 10: Kentucky counties in the national bottom 10% of BA+ share at each anchor year. The share dropped from 44.2% in 1940 to 24.2% in 2024.

Kentucky as a state has fallen further behind the U.S. average every decade, and that gap is now over 8 pp. But at the same time, the number of Kentucky counties in the true bottom tier of the national distribution has been cut nearly in half. Both things are true. The state average moves slowly because the state’s metros grew slower than metros nationally, but the state’s LOW end has substantially caught up. I think we cannot truly say that ”Kentucky is falling behind” and or ”Kentucky is catching up.” It is both, and which one we emphasize depends on whether we care about the state average or the shape of the state distribution.

File location

The trajectory table is at `output/tables/ky_vs_national_trajectory.csv`, the county gains table is at `output/tables/ky_county_gains_1940_2024.csv`, and the bottom-10% count is at `output/tables/ky_in_national_bottom10.csv`. The figure is at `output/figures/ky_vs_national_trajectory.png`. The build script is `code/29_kentucky_in_national_context.do`.

Data, Code, and Figures Availability

Everything used in this report, including the raw and cleaned data files, codes, and figures are available in the [GitHub repository](#). Anyone with access can rerun everything from the raw files through the final outputs and reproduce these results.